



Pymble Ladies' College

HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION 2014

Mathematics Extension 2

Time Allowed: 3 hours

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen.
Black pen is preferred.
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- In Questions 11-16, show relevant mathematical reasoning and/or calculations.
- Start each question in a **new** booklet.

Total Marks – 100

Section I Pages 1-4

10 marks

- Attempt all Questions 1-10
- Allow about 15 mins for this section

Section II Pages 5-12

90 marks

- Attempt Questions 11-16
- Allow about 2 hour 45 minutes for this section

Mark	/100
Highest Mark	/100
Rank	

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for Questions 1-10.

1 If $(a + bi)^2 = i$, then what are possible values for $a, b \in \mathbb{R}$?

(A) $a = \frac{1}{4}, b = \frac{1}{4}$

(B) $a = -\frac{1}{\sqrt{2}}, b = \frac{1}{\sqrt{2}}$

(C) $a = \frac{1}{\sqrt{2}}, b = \frac{1}{\sqrt{2}}$

(D) $a = \frac{1}{2}, b = \frac{1}{2}$

2 The polynomial $P(x) = x^3 + 3x^2 - 24x + 28$ has a double zero.

What is the value of the double zero?

(A) -7

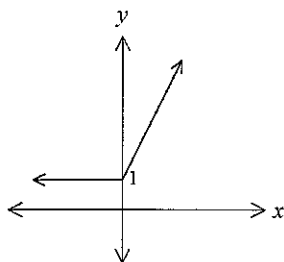
(B) -4

(C) 4

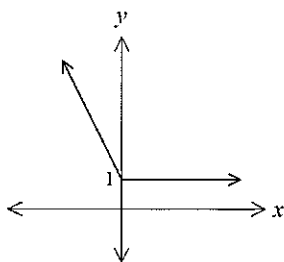
(D) 2

3 Which graph shows $y = 1 + x + |x|$?

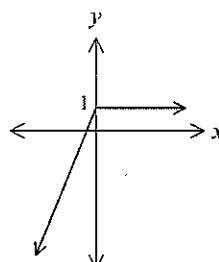
(A)



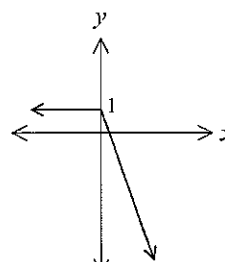
(B)



(C)



(D)



4 The graph of the ellipse $\frac{(x-1)^2}{9} + \frac{y^2}{4} = 1$ and the graph of the hyperbola $x^2 - y^2 = 4$ have

(A) no points in common.

(B) 1 point in common.

(C) 2 points in common.

(D) 3 points in common.

5 $\int \sin^{-1} 2x \, dx =$

(A) $x \sin^{-1} 2x + \frac{1}{4} \sqrt{1-4x^2} + C, \quad |x| \geq -1$

(B) $x \sin^{-1} 2x - \frac{1}{4} \sqrt{1-4x^2} + C, \quad |x| \geq -1$

(C) $x \sin^{-1} 2x + \frac{1}{2} \sqrt{1-4x^2} + C, \quad |x| \geq -1$

(D) $x \sin^{-1} 2x - \frac{1}{2} \sqrt{1-4x^2} + C, \quad |x| \geq -1$

6 Which of the following would be neither odd nor even?

(A) $y = x^2 \sin x$

(B) $y = \sin(x^2)$

(C) $y = (\sin x)^2$

(D) $y = x^2 + \sin x$

7 What is the exact value of $\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i\right)^{100}$?

(A) 1

(B) -1

(C) $\frac{1}{2^{50}}$

(D) $-\frac{1}{2^{50}}$

8 If $\frac{3x-19}{(x+3)(2x-1)} = \frac{a}{x+3} + \frac{b}{2x-1}$, then find the values of a and b .

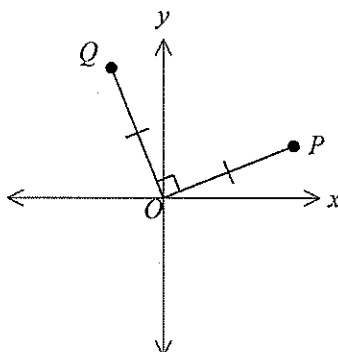
(A) $a = -4, \quad b = 5$

(B) $a = -4, \quad b = -5$

(C) $a = 4, \quad b = 5$

(D) $a = 4, \quad b = -5$

- 9 On the diagram P and Q represent complex numbers z and w respectively. Triangle OPQ is right angled and isosceles.



Which of the following is **false**?

- (A) $|z|^2 + |w|^2 = |z + w|^2$
- (B) $z^2 - w^2 = 0$
- (C) $z^2 + w^2 = 0$
- (D) $w = iz$
- 10 The ellipse $x^2 + 2ax + 2y^2 + 4by + 16 = 0$ has its centre at $(3, -2)$. Find the values of a and b .
- (A) $a = -3, b = -2$
- (B) $a = 2, b = -3$
- (C) $a = -3, b = 2$
- (D) $a = 3, b = 2$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section.

Answer each question in the appropriate writing booklet. Extra booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11. (15 marks). Use a **Separate Booklet**.

Marks

-
- (a) Use the substitution $u = 4 + \sin x$ to find

2

$$\int \frac{\sin x \cos x}{4 + \sin x} dx.$$

- (b) Let $w = -1 + \sqrt{3}i$ and $z = 1 - i$.

- (i) Find wz in the form $a + ib$.

1

- (ii) Find w and z in mod-arg form.

2

- (iii) Hence, find the exact value of $\sin \frac{5\pi}{12}$.

2

- (c) Let polynomial $P(x) = ax^6 - bx^5 + 1$.

- (i) State the conditions for α to be a zero of multiplicity two of $P(x)$.

1

- (ii) Given that $P(x)$ is divisible by $(x+1)^2$ find a and b .

3

- (d) The line $x = 1$ is a directrix and the point $(2, 0)$ is a focus of the conic whose eccentricity is $\sqrt{2}$.

- (i) Derive the equation of the conic.

3

- (ii) Prove that it is a rectangular hyperbola.

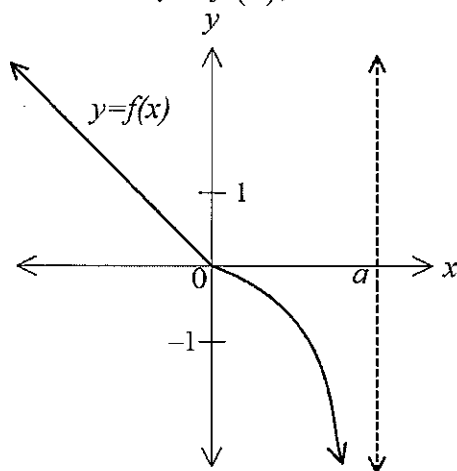
1

End of Question 11

(a) Find $\int \frac{\ln x}{x^2} dx$.

2

(b) The graph of the function $y = f(x)$, $x < a$ is shown below.



Sketch the following curves on separate half-page diagrams.

(i) $y = |f(x)|$.

1

(ii) $y = f(|x|)$.

1

(iii) $y = \frac{1}{f(x)}$.

2

(c) Let C be the curve $3e^{x-y} = x^2 + y^2 + 1$.

3

Find the equation of the tangent to C at the point $(1,1)$.

Question 12 continues on page 7

Question 12 (continued).**Marks**

-
- | | | | |
|-----|-------|--|---|
| (d) | (i) | Expand $(a-b)^3$. | 1 |
| | (ii) | Solve $z^3 = -1$. | 2 |
| | (iii) | Express the polynomial $z^3 - 3iz^2 - 3z + 1 + i$ in the form $(z+p)^3 + q$ where p is an imaginary number and q is a real number. | 1 |
| | (iv) | Hence solve $z^3 - 3iz^2 - 3z + 1 + i = 0$ giving the solution in the form $z = x + iy$ where $x, y \in \mathbb{R}$. | 2 |

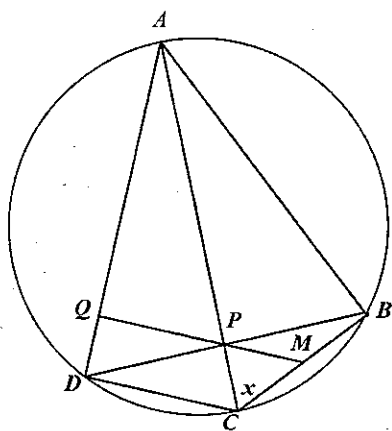
End of Question 12

- (a) When the polynomial $p(x) = x^4 + ax + 2$ is divided by $x^2 + 1$ the remainder is $2x + 3$.
Find the value of a . 2

- (b) Using the substitution $t = \tan \frac{x}{2}$, find $\int \frac{\tan x}{1 + \cos x} dx$. 3

- (c) Consider the region bounded by the curve $y = x^2 - 6x + 8$ and the x -axis.
Use the method of cylindrical shells to find the volume of the solid formed if the region is rotated about the y -axis to form a solid of revolution. 3

- (d) $ABCD$ is a cyclic quadrilateral. Diagonals AC and BD intersect at right angles at P .
 M is the midpoint of BC . MP produced meets AD at Q . Let $\angle MCP = x$.

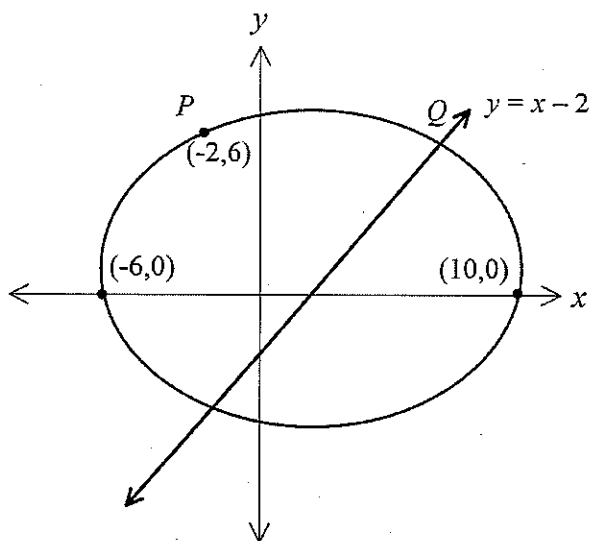


- (i) Show $\angle MCP = \angle CPM$. 2
- (ii) Show $MQ \perp AD$. 2

Question 13 continues on page 9

- (e) The ellipse shown below passes through point $P(-2, 6)$.
The centre of the ellipse lies on the x -axis, and the ellipse passes through the points $(-6, 0)$ and $(10, 0)$.

3



The line shown is $y = x - 2$. This line intersects the ellipse at Q .

What is the x coordinate of point Q ?

End of Question 13

(a) (i) Derive the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at point $P(x_1, y_1)$. 3

(ii) The tangents to the ellipse $x^2 + 4y^2 = 4$ at the points $P(2\cos\theta, \sin\theta)$ and $Q(2\cos\phi, \sin\phi)$ are at right angles to each other. 2

Show that $4\tan\theta\tan\phi = -1$.

(b) If w is one of the complex roots of $z^3 = 1$, simplify 3
 $(1-w)(1-w^2)(1-w^4)(1-w^8)$.

(c) (i) If $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$ prove that $I_n + I_{n-2} = \frac{1}{n-1}$, $n > 2$. 2

(ii) Hence, evaluate $\int_0^{\frac{\pi}{4}} \tan^5 x \, dx$. 2

(d) Sketch the locus of z if $\frac{z-2i}{z-1}$ is purely imaginary. 3

End of Question 14

- (a) The base of a solid is given by the region in the xy -plane enclosed by the curve $y = x^2$ and $y = 8 - x^2$.
Each cross-section perpendicular to the x -axis is a square.
- (i) Show that the area of the square cross-section at $x = h$ is $(8 - 2h^2)^2$. 1
- (ii) Hence, find the volume of the solid. 3
- (b) Show that $\int_0^1 \frac{dx}{x^2 - x + 1} = \frac{2\sqrt{3}\pi}{9}$. 2
- (c) Let $f(x) = \frac{4}{x-1} - \frac{4}{x+1} - 1$, where $x \neq \pm 1$.
- (i) Find the x and y intercepts of the graph of $y = f(x)$. 2
- (ii) Show that $y = f(x)$ is an even function. 1
- (iii) Find the equation of the horizontal asymptote. 1
- (iv) Sketch the graph of $y = f(x)$. 2
- (v) Let S be the area bound by the graph of $y = f(x)$, the straight lines $x = 3$, $x = a$ ($a > 3$) and $y = -1$. 3

Find S in terms of a and deduce that $S < 4 \ln 2$.

End of Question 15

- (a) The locus of w is described by the equation $|w+3| = |w-2+5i|$.
- (i) Sketch on an Argand Diagram the locus of w . 2
- (ii) Find the Cartesian equation of the locus of w . 2
- (b) (i) Given that $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$ explain why $\frac{1}{(n+1)^2} < \frac{1}{n(n+1)}, n \in \mathbb{Z}^+$. 1
- (ii) Using induction, prove $S_n = \sum_{r=1}^n \frac{1}{r^2} \leq 2 - \frac{1}{n}, n \geq 1$. 3
- (c) Consider the quadratic equation $x^2 - x + k = 0$ where k is a real number. The equation has 2 distinct positive roots α and β .
- (i) Show $0 < k < \frac{1}{4}$. 2
- (ii) Show that $\frac{1}{\alpha^2} + \frac{1}{\beta^2} > 8$. 2
- (d) Given $I = \int_{-1}^1 \frac{x^2 e^x}{e^x + 1} dx$ and $J = \int_{-1}^1 \frac{x^2}{e^x + 1} dx$.
- (i) Use the substitution $u = -x$ in I to show $I = J$. 1
- (ii) Hence evaluate I and J . 2

End of Paper

①

Solutions

1. $a^2 - b^2 + 2abi$

$a^2 - b^2 = 0$

$2ab = 1$

$ab = \frac{1}{2}$

∴ C

2. $P(x) = x^3 + 3x^2 - 24x + 28$

$P'(x) = 3x^2 + 6x - 24$

When $P'(x) = 0$

$x^2 + 2x - 8 = 0$

$(x+4)(x-2) = 0$

∴ $x = 2, -4$

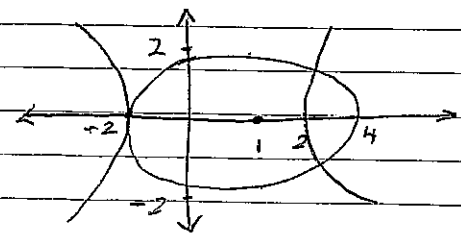
$P(2) = 8 + 12 - 48 + 28$

$= 0$

∴ D

3. A.

4. $\frac{(x-1)^2}{9} + \frac{y^2}{4} = 1$ D



5. $\int \sin^{-1} 2x \, dx =$ $u = \sin^{-1} 2x$ $dv = 1$
 $du = \frac{2}{\sqrt{1-4x^2}} dx$ $v = x$

$= x \sin^{-1} 2x - \int \frac{2x}{\sqrt{1-4x^2}} dx$

$= x \sin^{-1} 2x + \frac{1}{2} \sqrt{1-4x^2} + C$

∴ C

6. D

7. $\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} i \right)^{100} = \left(\text{cis} \left(-\frac{\pi}{4} \right) \right)^{100}$
 $= \text{cis} (-25\pi)$

$= \text{cis } \pi$

$= -1$

8. D.

9. B.

10. C

(2)

Question 11

$$a) \int \frac{\sin x \cos x}{4 + \sin x} dx \quad u = 4 + \sin x$$

$$du = \cos x dx$$

$$= \int \frac{(u-4) du}{u}$$

$$= \int \left(1 - \frac{4}{u}\right) du$$

$$= u - 4 \ln|u| + C$$

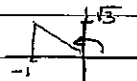
$$= 4 + \sin x - 4 \ln|4 + \sin x| + C$$

$$b) i) w = -1 + \sqrt{3}i \quad z = 1 - i$$

$$wz = (-1 + \sqrt{3}i)(1 - i)$$

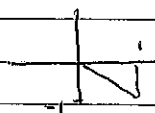
$$= -1 + i + \sqrt{3}i + \sqrt{3}$$

$$= \sqrt{3} - 1 + (1 + \sqrt{3})i$$

$$ii) w = 2 \operatorname{cis} \frac{2\pi}{3}$$


$$\arg = \tan^{-1}(\sqrt{3}) = \frac{2\pi}{3}$$

$$z = \sqrt{2} \operatorname{cis}(-\pi/4)$$



$$iii) \sin \frac{5\pi}{12}$$

$$wz = 2\sqrt{2} \operatorname{cis}\left(\frac{2\pi}{3} - \frac{\pi}{4}\right)$$

$$= 2\sqrt{2} \operatorname{cis} \frac{5\pi}{12}$$

$$wz = \sqrt{3} - 1 + (1 + \sqrt{3})i \quad \text{from i)}$$

$$\therefore \sqrt{3} - 1 + (1 + \sqrt{3})i = 2\sqrt{2} \cos \frac{5\pi}{12} + 2\sqrt{2}i \sin \frac{5\pi}{12}$$

Equating parts

$$2\sqrt{2} \sin \frac{5\pi}{12} = 1 + \sqrt{3}$$

$$\therefore \sin \frac{5\pi}{12} = \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

$$c) P(x) = ax^6 - bx^5 + 1$$

$$i) P(x) = 0 = P'(x)$$

$$ii) P(-1) = a + b + 1 = 0 \quad -①$$

$$P'(x) = 6ax^5 - 5bx^4$$

$$P'(-1) = -6a - 5b = 0 \quad -②$$

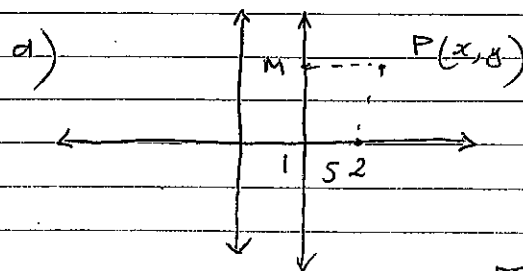
$$5a + 5b = -5$$

$$-6a - 5b = 0$$

$$-a = -5$$

$$a = 5$$

$$b = -6$$



Let $P(x, y)$ be a point on the conic

Let S be the focus

M be the foot of the perpendicular at the directrix

$$\frac{PS}{PM} = \sqrt{2}$$

$$d_{PS} = \sqrt{(x-2)^2 + y^2}$$

$$d_{PM} = \sqrt{(x-1)^2}$$

$$\frac{d_{PS}}{d_{PM}} = \sqrt{2}$$

$$\frac{\sqrt{(x-2)^2 + y^2}}{\sqrt{(x-1)^2}} = \sqrt{2}$$

$$\frac{(x-2)^2 + y^2}{(x-1)^2} = 2$$

$$(x-1)^2$$

$$x^2 - 4x + 4 + y^2 = 2x^2 - 4x + 2$$

$$x^2 - y^2 = 2 \quad \frac{x^2}{2} - \frac{y^2}{2} = 1$$

ii) $e = \sqrt{2} \therefore$ hyperbola is rectangular.

$$\frac{x^2}{2} - \frac{y^2}{2} = 1$$

Gradients of asymptotes $\frac{\sqrt{2}}{\sqrt{2}} x - \frac{\sqrt{2}}{\sqrt{2}}$

$$= -1$$

$\therefore$ Asymptotes are perpendicular
 $\therefore$ Rectangular

Question 12

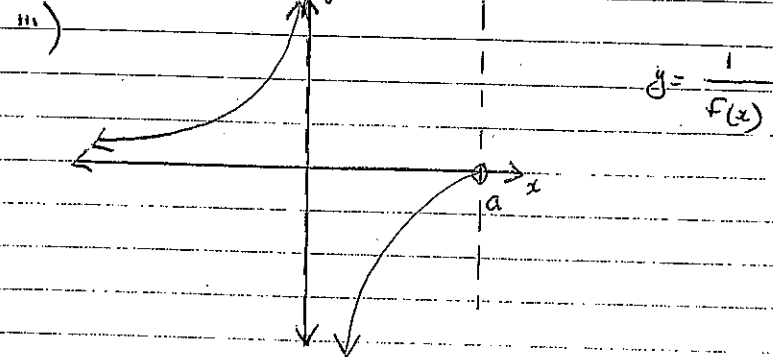
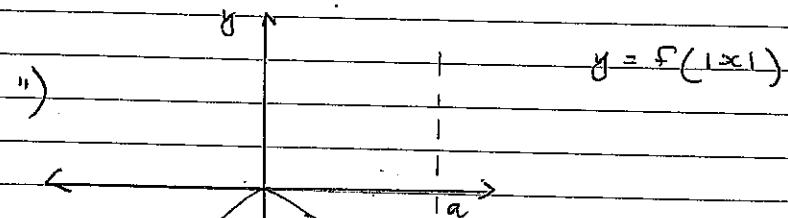
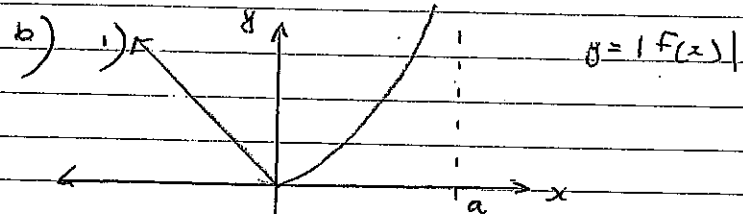
a) $\int \frac{\ln x}{x^2} dx$

$$u = \ln x \quad dv = x^{-2}$$

$$du = \frac{1}{x} \quad v = -x^{-1}$$

$$= -\frac{\ln x}{x} + \int x^{-2} dx$$

$$= -\frac{\ln x}{x} - \frac{1}{x} + C$$



$$c) \quad 3e^{x-y} = x^2 + y^2 + 1$$

$$3e^{x-y} \times \left(1 - \frac{dy}{dx}\right) = 2x + 2y \frac{dy}{dx}$$

$$3e^{x-y} - \frac{dy}{dx} \times 3e^{x-y} - 2y \frac{dy}{dx} = 2x$$

$$-\frac{dy}{dx} (3e^{x-y} + 2y) = 2x - 3e^{x-y}$$

$$\frac{dy}{dx} = \frac{3e^{x-y} - 2x}{3e^{x-y} + 2y}$$

$$\text{When } x=1, y=1 \quad \frac{dy}{dx} = \frac{3e^0 - 2}{3e^0 + 2} = \frac{1}{5}$$

$$\text{Eq}^n \text{ of tangent } y-1 = \frac{1}{5}(x-1)$$

$$5y-5 = x-1$$

$$x-5y+4=0$$

$$d) \quad 1) \quad (a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$ii) \quad z^3 = -1$$

$$z^3 + 1 = 0$$

$$(z+1)(z^2 - z + 1) = 0$$

$$\therefore z = -1 \quad z = \frac{1 \pm \sqrt{1-4}}{2}$$

$$= \frac{1 \pm \sqrt{3}i}{2}$$

$$iii) \quad z^3 - 3iz^2 - 3z + 1 + i = (z-i)^3 + 1$$

$$iv) \quad z^3 - 3iz^2 - 3z + 1 + i = 0$$

$$(z-i)^3 + 1 = 0$$

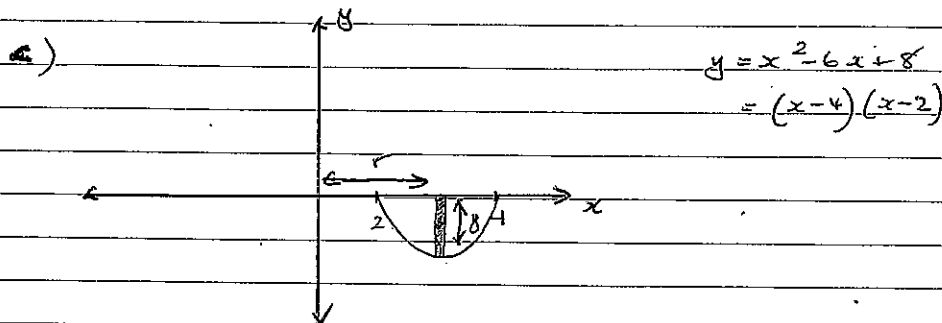
$$(z-i+1)((z-i)^2 - (z-i) + 1) = 0$$

$$\therefore z-i = -1 \quad \text{or} \quad z-i = \frac{1 \pm \sqrt{3}i}{2}$$

$$z = i-1$$

$$z = i + \frac{1 \pm \sqrt{3}i}{2}$$

Question 13



$$\text{Thickness} = dx$$

$$r = x$$

$$V_{\text{slice}} = 2\pi r h dx$$

$$h = y = |x^2 - 6x + 8|$$

$$dV = \sum_2^4 2\pi x |x^2 - 6x + 8| dx$$

$$V = \lim_{dx \rightarrow 0} \sum_2^4 2\pi x |x^2 - 6x + 8| dx$$

$$= \left| 2\pi \int_2^4 (x^3 - 6x^2 + 8x) dx \right|$$

$$= \left| 2\pi \left[\frac{x^4}{4} - \frac{6x^3}{3} + \frac{8x^2}{2} \right]_2^4 \right|$$

$$= \left| 2\pi \left\{ \frac{4^4}{4} - 2 \times 4^3 + 4 \times 4^2 - \left(\frac{2^4}{4} - 2 \times 2^3 + 4 \times 2^2 \right) \right\} \right|$$

$$= \left| 2\pi \{ 64 - 128 + 64 - 4 + 16 - 16 \} \right|$$

$$= 8\pi \text{ units}^3$$

b) $t = \tan \frac{x}{2}$ $dx = \frac{2}{1+t^2} dt$ $\int \frac{\tan x}{1+\cos x} dx$

$\frac{x}{2} = \tan^{-1} t$
 $x = 2 \tan^{-1} t$

$\tan x = \frac{2t}{1-t^2}$

$\frac{1+t^2}{1-t^2} \cdot 2t$

$= \int \frac{\frac{2t}{1-t^2}}{1 + \frac{1-t^2}{1+t^2}} \times \frac{2}{1+t^2} dt$

$= \int \frac{\frac{2t}{1-t^2}}{\frac{1+t^2+1-t^2}{1+t^2}} \times \frac{2}{1+t^2} dt$

$= \int \frac{2t}{1-t^2} \times \frac{1+t^2}{2} \times \frac{2}{1+t^2} dt$

$= \int \frac{2t}{1-t^2} dt$

$= -\ln |1-t^2| + C$

$= -\ln |1-\tan^2 \frac{x}{2}| + C$

8) $P(x) = x^4 + ax + 2$

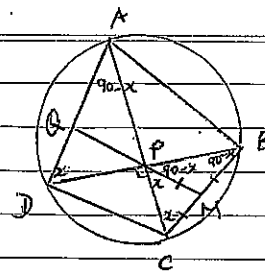
$P(x) = (x^2+1)A(x) + 2x+3$

$P(x) = (x^2+1)(x^2-1) + 2x+3$

$= x^4 - 1 + 2x + 3$

$= x^4 + 2x + 2$

$\therefore a = 2$



d) 1) $\angle BPC = 90^\circ$
 $\therefore BC$ is a diameter of a circle passing through BPC, M is the midpoint of BC
 $\therefore M$ is the centre of this circle.

$\therefore PM = MC$ radii of this circle

$\therefore \triangle PMC$ is isosceles

$\therefore \angle MCP = \angle CPM = x$, equal angles are opposite equal sides in an isosceles triangle.

ii) $\angle CPM = x$ (from i))

$\angle CPD = 90^\circ$ (given)

$\angle QPD = 90 - x$ (since QM is a straight line)

$\angle ADB = x$ (angles at the circumference standing on the same arc AB are equal)

$\therefore \angle PQD = 90^\circ$ (angle sum $\triangle PQD$)

$$e) \quad 2a = 16 \\ a = 8$$

Eqⁿ of ellipse

$$\frac{(x-2)^2}{64} + \frac{y^2}{b^2} = 1$$

Passes through $(-2, 6)$

$$\therefore \frac{(-2-2)^2}{64} + \frac{6^2}{b^2} = 1$$

$$\frac{16}{64} + \frac{36}{b^2} = 1$$

$$\frac{36}{b^2} = \frac{48}{64}$$

$$b^2 = \frac{64 \times 36}{48}$$

$$b^2 = 48$$

$$\therefore \frac{(x-2)^2}{64} + \frac{y^2}{48} = 1 \quad y = x - 2$$

$$\frac{(x-2)^2}{64} + \frac{(x-2)^2}{48} = 1$$

$$48(x^2 - 4x + 4) + 64(x^2 - 4x + 4) = 3072$$

$$112x^2 - 448x + 448 = 3072$$

$$x^2 - 4x + 4 = \frac{3072}{112}$$

$$(x-2)^2 = \frac{3072}{112} \quad x = 2 + \frac{8\sqrt{31}}{7}$$

$$x - 2 = \pm \frac{\sqrt{3072}}{\sqrt{112}} \quad \text{is the } x$$

$$x = 2 \pm \frac{32\sqrt{3}}{4\sqrt{7}}$$

$$= 2 \pm \frac{8\sqrt{3} \times \sqrt{7}}{1}$$

Question 14

$$a) \quad 1) \quad \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \times \frac{b^2}{2y} \\ = -\frac{b^2 x}{a^2 y}$$

$$At (x_1, y_1) \quad \frac{dy}{dx} = -\frac{b^2 x_1}{a^2 y_1}$$

Eqⁿ of tangent

$$y - y_1 = -\frac{b^2 x_1}{a^2 y_1} (x - x_1)$$

$$a^2 y_1 y - a^2 y_1^2 = -b^2 x_1 x + b^2 x_1^2$$

$$a^2 y_1 y + b^2 x_1 x = a^2 y_1^2 + b^2 x_1^2$$

$$ii) \quad x^2 + 4y^2 = 4$$

$$\frac{x^2}{4} + y^2 = 1 \quad P(2\cos\theta, \sin\theta) \quad Q(2\cos\phi, \sin\phi)$$

$$\text{From i) } y = \frac{-b^2 x_1}{a^2 y_1} x + \frac{a^2 y_1^2}{a^2 y_1} + \frac{b^2 x_1^2}{a^2 y_1}$$

$$\therefore m_p = \frac{-b^2 \times 2\cos\theta}{a^2 \sin\theta}$$

$$m_Q = \frac{-b^2 \times 2\cos\phi}{a^2 \sin\phi}$$

$m_p \times m_Q = -1$ since tangent at P is perpendicular to tangent at Q

$$\therefore \frac{-b^2 \times 2\cos\theta}{a^2 \sin\theta} \times \frac{-b^2 \times 2\cos\phi}{a^2 \sin\phi} = -1$$

$$\frac{4b^4 \cos \theta \cos \phi}{a^4 \sin \theta \sin \phi} = -1$$

$$a^2 = 4, \quad b^2 = 1$$

$$\therefore \frac{4 \times 1^2 \cos \theta \cos \phi}{4^2 \sin \theta \sin \phi} = -1$$

$$\cos \theta \cos \phi = -4 \sin \theta \sin \phi$$

$$1 = \frac{-4 \sin \theta \sin \phi}{\cos \theta \cos \phi}$$

$$4 \tan \theta \tan \phi = -1$$

$$b) \quad z^3 = 1$$

$$z^3 - 1 = 0$$

$$(z-1)(z^2 + z + 1) = 0$$

$$\therefore z = 1 \quad z = \frac{-1 \pm \sqrt{-3}}{2}$$

$$z = \frac{-1 \pm \sqrt{3}i}{2}$$

$$z_1 = 1$$

$$z_2 = \text{cis } \frac{2\pi}{3}$$

$$z_3 = \text{cis } (-\frac{2\pi}{3})$$

$$\text{Let } w = \text{cis } \frac{2\pi}{3}$$

$$w^2 = \text{cis } \frac{4\pi}{3} = \text{cis } (-\frac{2\pi}{3})$$

$$w^4 = \text{cis } \frac{8\pi}{3} = \text{cis } \frac{2\pi}{3}$$

$$w^8 = \text{cis } \frac{16\pi}{3} = \text{cis } (-\frac{2\pi}{3})$$

$$\therefore (1 - \text{cis } \frac{2\pi}{3})(1 - \text{cis } (-\frac{2\pi}{3}))(1 - \text{cis } \frac{2\pi}{3})(1 - \text{cis } (-\frac{2\pi}{3}))$$

$$= (1 - \text{cis } \frac{2\pi}{3})^2 (1 - \text{cis } (-\frac{2\pi}{3}))^2$$

$$= (1 - (-\frac{1}{2} + \frac{\sqrt{3}}{2}i))^2 (1 - (-\frac{1}{2} - \frac{\sqrt{3}}{2}i))^2$$

$$= (\frac{3}{2} - \frac{\sqrt{3}}{2}i)^2 (\frac{3}{2} + \frac{\sqrt{3}}{2}i)^2$$

$$= ((\frac{3}{2} - \frac{\sqrt{3}}{2}i)(\frac{3}{2} + \frac{\sqrt{3}}{2}i))^2 = (\frac{9}{4} + \frac{3}{4})^2 = 9$$

$$c) \quad 1) \quad I_n = \int_0^{\pi/4} \tan^n x \, dx$$

$$I_{n-2} = \int_0^{\pi/4} \tan^{n-2} x \, dx$$

$$I_n + I_{n-2} = \int_0^{\pi/4} \tan^n x + \tan^{n-2} x \, dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x (\tan^2 x + 1) \, dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx \quad \begin{matrix} u = \tan x \\ du = (n-2) \tan^{n-3} x \, dx \\ v = \tan x \\ \sec^2 x \, dx \end{matrix}$$

$$= \tan^{n-2} x \tan x \Big|_0^{\pi/4} - \int_0^{\pi/4} \tan x \times (n-2) \tan^{n-3} x \sec^2 x \, dx$$

$$= \tan^{n-1} x \Big|_0^{\pi/4} - (n-2) \int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx$$

$$= 1 - (n-2) \int_0^{\pi/4} \tan^n x + \tan^{n-2} x \, dx$$

$$\frac{I_n + I_{n-2}}{1-2} = 1 - (n-2) I_n - (n-2) I_{n-2}$$

$$I_n + I_{n-2} = 1 - n I_n + 2 I_n - n I_{n-2} + 2 I_{n-2}$$

$$I_n + n I_n - 2 I_n + I_{n-2} + n I_{n-2} - 2 I_{n-2} = 1$$

$$(n-1) I_n + (n-1) I_{n-2} = 1$$

$$\frac{I_n + I_{n-2}}{n-1} = \frac{1}{n-1} \quad \text{As req'd}$$

$$11) \int_0^{\pi/4} \tan^5 x \, dx = I_5$$

$$I_5 + I_3 = \frac{1}{4}$$

$$I_3 + I_1 = \frac{1}{2}$$

$$I_1 = \int_0^{\pi/4} \tan x \, dx$$

$$= -\ln |\cos x| \Big|_0^{\pi/4}$$

$$= -\ln |\cos \pi/4| + \ln |\cos 0|$$

$$= -\ln \frac{1}{\sqrt{2}}$$

$$= \ln \sqrt{2}$$

$$I_3 + \ln \sqrt{2} = \frac{1}{2}$$

$$I_3 = \frac{1}{2} - \frac{1}{2} \ln 2$$

$$I_5 + \frac{1}{2} - \frac{1}{2} \ln 2 = \frac{1}{4}$$

$$I_5 = \frac{1}{2} \ln 2 - \frac{1}{4}$$

$$d) \frac{z-2i}{z-1} \quad \text{let } z = x+iy$$

$$\frac{x+iy-2i}{x+iy-1}$$

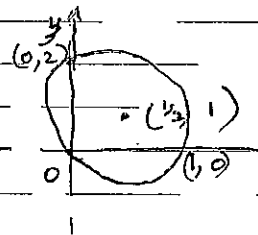
$$= \frac{x+(y-2)i}{x-1+iy} \times \frac{x-1-iy}{x-1-iy}$$

$$= \frac{x^2 - x - 2xy + x(y-2)i - (y-2)i - (y-2)i + y(y-2)}{(x-1)^2 + y^2}$$

If locus is imaginary then

$$\frac{x^2 - x + y^2 - 2y}{(x-1)^2 + y^2} = 0$$

$$\therefore (x - 1/2)^2 + (y-1)^2 = 5/4$$



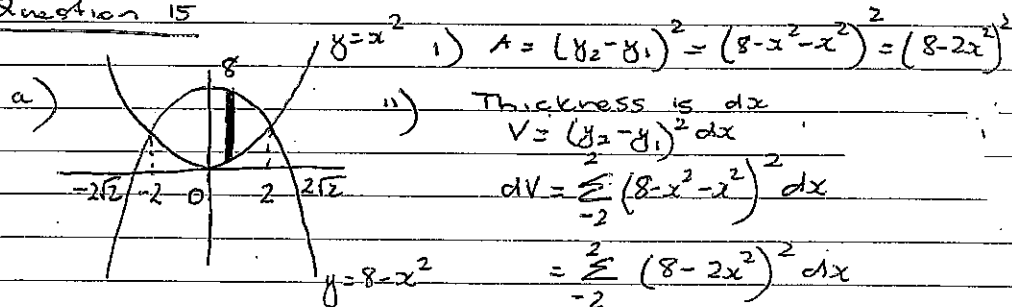
When $x=0$

$$(y-1)^2 = 1$$

$$y-1 = \pm 1$$

$$y = 2, 0$$

Question 15



$$y = x^2$$

$$y = 8 - x^2$$

$$x^2 = 8 - x^2$$

$$2x^2 = 8$$

$$x^2 = 4$$

$$x = \pm 2$$

$$V = \lim_{dx \rightarrow 0} 2 \int_0^2 (8 - 2x^2)^2 dx$$

$$= 8 \int_0^2 (4 - x^2)^2 dx$$

$$= 8 \int_0^2 16 - 8x^2 + x^4 dx$$

$$= 8 \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_0^2$$

$$= 8 \left\{ 16 \times 2 - \frac{8 \times 2^3}{3} + \frac{2^5}{5} - 0 \right\}$$

$$= \frac{7168}{15} \text{ units}^3$$

$$b) \int_0^1 \frac{dx}{x^2 - x + 1}$$

$$= \int_0^1 \frac{dx}{x^2 - x + \frac{1}{4} + \frac{3}{4}}$$

$$= \int_0^1 \frac{dx}{(x - \frac{1}{2})^2 + \frac{3}{4}}$$

$$= \left[\frac{2}{\sqrt{3}} \tan^{-1} \frac{2(x - \frac{1}{2})}{\sqrt{3}} \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \frac{2(1 - \frac{1}{2})}{\sqrt{3}} - \frac{2}{\sqrt{3}} \tan^{-1} \frac{2(0 - \frac{1}{2})}{\sqrt{3}}$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}} - \frac{2}{\sqrt{3}} \tan^{-1} \left(-\frac{1}{\sqrt{3}} \right)$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}}$$

$$= \frac{4}{\sqrt{3}} \times \frac{\pi}{6}$$

$$= \frac{2\pi}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{2\sqrt{3}\pi}{9} \text{ As req'd}$$

$$c) f(x) = \frac{4}{x-1} - \frac{4}{x+1} - 1 \quad x \neq \pm 1$$

$$i) f(0) = \frac{4}{0-1} - \frac{4}{0+1} - 1 \quad \text{when } f(x) = 0$$

$$= -4 - 4 - 1 = -9$$

$$\frac{4}{x-1} - \frac{4}{x+1} - 1 = 0$$

$$\frac{4(x+1) - 4(x-1)}{(x^2-1)} = 1$$

$$4x+4-4x+4 = x^2-1$$

$$x^2 - 9 = 0$$

$$x = \pm 3$$

$$ii) f(a) = \frac{4}{a-1} - \frac{4}{a+1} - 1$$

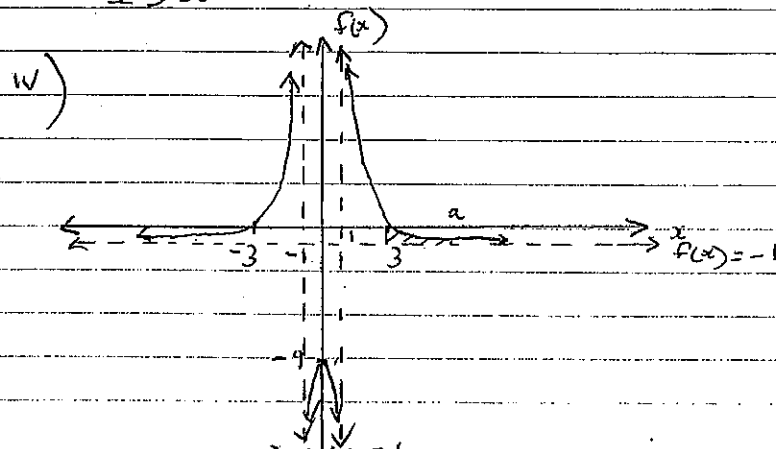
$$f(-a) = \frac{4}{-a-1} - \frac{4}{-a+1} - 1$$

$$= -\frac{4}{a+1} + \frac{4}{a-1} - 1$$

$$= -\frac{4}{a-1} - \frac{4}{a+1} - 1$$

Since $f(a) = f(-a)$ $f(x)$ is an even fⁿ

$$iii) \lim_{x \rightarrow \infty} f(x) = -1$$



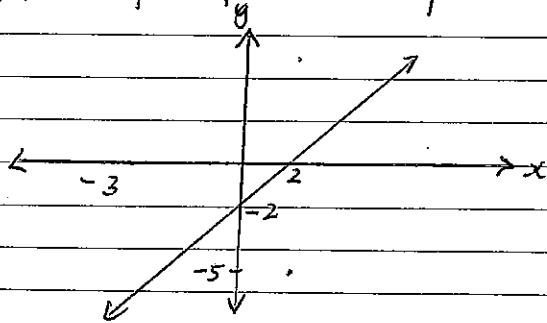
$$\begin{aligned}
 \text{v) } S &= \int_3^a \frac{4}{x-1} - \frac{4}{x+1} - 1 \, dx \\
 &= \left[4 \ln|x-1| - 4 \ln|x+1| - x \right]_3^a \\
 &= \{ 4 \ln|a-1| - 4 \ln|a+1| - 1 - (4 \ln 2 - 4 \ln 4 - 1) \} \\
 &= 4 \ln \left| \frac{a-1}{a+1} \right| - 1 - 4 \ln \frac{1}{2} + 1 \\
 &= 4 \ln \left| \left(\frac{a-1}{a+1} \right) \right| + 4 \ln 2
 \end{aligned}$$

$$\text{Since } a > 3 \quad \frac{a-1}{a+1} < 1 \quad \therefore \ln \left| \frac{a-1}{a+1} \right| < 0$$

$$\therefore S < 4 \ln 2$$

Q16

$$\text{a) i) } |w+3| = |w-2+5i|$$



$$\text{Let } w = x + iy$$

$$\text{ii) } |x+3+iy| = |x-2+i(y+5)|$$

$$\sqrt{(x+3)^2 + y^2} = \sqrt{(x-2)^2 + (y+5)^2}$$

$$x^2 + 6x + 9 + y^2 = x^2 - 4x + 4 + y^2 + 10y + 25$$

$$10x - 10y - 20 = 0$$

$$x - y - 2 = 0$$

$$\text{b) i) } \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$$

Multiply b.s by $\frac{1}{n+1}$

$$\frac{1}{n(n+1)} - \frac{1}{(n+1)^2} = \frac{1}{n(n+1)^2}$$

$$\frac{1}{(n+1)^2} = \frac{1}{n(n+1)} - \frac{1}{n(n+1)^2}$$

$$\text{Since } n \in \mathbb{Z}^+ \quad \frac{1}{n(n+1)^2} > 0$$

$$\therefore \frac{1}{(n+1)^2} < \frac{1}{n(n+1)}$$

Let the statement be

$$\text{ii) } S_n = \sum_{r=1}^n \frac{1}{r^2} \leq 2 - \frac{1}{n}, \quad n \geq 1$$

When $n=1$

$$\text{LHS } S_1 = 1 \quad \text{RHS} = 2 - \frac{1}{1}$$

Statement

$\therefore$ True for $n=1$

Assume the statement is true for $n=k$

$$\text{i.e. } S_k = \sum_{r=1}^k \frac{1}{r^2} \leq 2 - \frac{1}{k}; \quad k \geq 1$$

RTP true for $n=k+1$

$$\text{i.e. } S_{k+1} = \sum_{r=1}^{k+1} \frac{1}{r^2} \leq 2 - \frac{1}{k+1}; \quad k \geq 1$$

$$\text{Now } S_{k+1} = S_k + T_{k+1}$$

$$= \sum_{r=1}^k \frac{1}{r^2} + \frac{1}{(k+1)^2}$$

$$\leq 2 - \frac{1}{k} + \frac{1}{(k+1)^2}$$

$$\leq 2 - \frac{1}{k} + \frac{1}{k(k+1)} \quad (\text{from 1})$$

$$\leq 2 - \frac{k-1+1}{k(k+1)}$$

$$\leq 2 - \frac{k}{k(k+1)}$$

$$\leq 2 - \frac{1}{k+1} \quad k \geq 1$$

Hence since the statement is true for $n=1$ and $n=k$ and $n=k+1$, the statement is true for $n=1+1=2$ and so on by the principle of mathematical induction.

Q 16

$$(c) (i) \quad x^2 - x + k = 0$$

Since 2 distinct positive roots

$$b^2 - 4ac > 0$$

$$(-1)^2 - 4 \times 1 \times k > 0$$

$$1 - 4k > 0 \quad (1)$$

$$4k < 1$$

$$k < \frac{1}{4}$$

Product of roots = $+k > 0$ (since both roots positive)

$$\therefore 0 < k < \frac{1}{4}$$

$$\frac{\alpha^2}{\beta^2}$$

$$\text{i.e. } \frac{1}{\alpha^2} + \frac{1}{\beta^2} - 8 > 0$$

$$\frac{\alpha^2 + \beta^2 - 8\alpha^2\beta^2}{\alpha^2\beta^2} > 0$$

Since

numerator > 0

since $\alpha^2\beta^2 > 0$.

$$\text{Now } \alpha^2 + \beta^2 - 8\alpha^2\beta^2$$

$$= (\alpha + \beta)^2 - 2\alpha\beta - 8\alpha^2\beta^2$$

$$= (1)^2 - 2k - 8k^2 \quad (\alpha\beta = k, \alpha + \beta = 1)$$

$$= 1 - 2k - 8k^2$$

$$= (1 - 4k)(1 + 2k)$$

$$\text{From (1) } 1 - 4k > 0 \quad \text{and } 1 + 2k > 0 \quad (k > 0)$$

$$1 - 2k - 8k^2 > 0$$

$$\text{and } \frac{1}{\alpha^2} + \frac{1}{\beta^2} > 8$$

$$(d) (i) \quad I = \int_{-1}^1 \frac{x^2 e^x}{e^{x^2} + 1} dx$$

$$J = \int_{-1}^1 \frac{x^2}{e^{x^2} + 1} dx$$

$$\text{Let } u = -x \quad \frac{du}{dx} = -1$$

$$\text{When } x = -1 \quad u = 1$$

$$x = 1 \quad u = -1$$

$$\therefore I = \int_{-1}^1 \frac{u^2 e^{-u}}{e^{-u} + 1} - du$$

$$= \int_{-1}^1 \frac{u^2 e^{-u}}{e^{-u} + 1} du$$

$$= \int_{-1}^1 \frac{u^2 e^{-u}}{e^{-u} + 1} \times \frac{e^u}{e^u} du$$

$$= \int_{-1}^1 \frac{u^2}{1 + e^u} du = J$$

$$(ii) \quad I + J = \int_{-1}^1 \frac{x^2 e^x + x^2}{e^x + 1} dx$$

$$= \int_{-1}^1 \frac{x^2 (e^x + 1)}{(e^x + 1)} dx$$

$$= \left[\frac{x^3}{3} \right]_{-1}^1$$

$$= \frac{1}{3} - \frac{-1}{3} = \frac{2}{3} = 2I = 2J$$

$$\therefore I = J = \frac{1}{3}$$